Use of sign languages by the general hearing population was more widespread at an earlier stage in the evolution of language: Evidence from Kode Papua

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Previous studies have suggested that in cases where the language landscapes of esoteric societies differ in systematic ways from those of their exoteric counterparts, the linguistic features associated with esoteric societies may be projected back to an earlier stage in the evolution of language. This paper focuses on one such feature of a sociolinguistic nature, namely the widespread use of sign languages by the general hearing population. A striking case study is provided by Kode Papua, a hitherto-undescribed sign language used in parts of western New Guinea; moreover, the use of sign languages by general hearing populations may be widespread in other parts of the world typically associated with esoteric societies. The existence of Kode Papua alongside potentially similar cases in other esoteric societies thus suggests that the use of sign languages by general hearing populations may have been more common at an earlier stage in the evolution of language.

1. Introduction

Recent research by several scholars has provided substantial empirical support for various claims to the effect that sociopolitical features drive linguistic structures (Dahl 2004, Wray and Grace 2007, Lupyan and Dale 2010, Trudgill 2011, Raviv et al 2020, Chen et al 2024 and many others). More specifically, several of these studies suggest that languages spoken in larger, open, and more *exoteric* societies, such as those that have been characterized as WEIRD (Western Educated Industrial Rich Democratic), differ in systematic ways from their counterparts spoken in smaller, close-knit, and more *esoteric* societies, many of which tend to be associated with the so-called "Global South".

Although concerned with contemporary human languages, such studies have a major bearing on the evolution of language. Until relatively recently, all human societies were of an esoteric nature. Thus, given the connection between sociopolitical features and linguistic structures, it has been suggested that linguistic features associated with contemporary esoteric societies are plausible

candidates for reconstruction to a prior stage in the evolution of language (Gil 2021, Benítez-Burraco et al 2024, Chen et al 2024).

Invoking this method of inference into our linguistic past, this paper focuses on one particular aspect of the sociolinguistic profile of esoteric societies, namely, the common use of sign languages by hearing people. Accordingly, this paper argues that the widespread use of sign languages by the general hearing population should be reconstructed to an earlier stage in the evolution of language.

When the field of sign language studies emerged in the previous century, its original focus was on the national sign languages of major, mostly WEIRD societies, such as ASL (American Sign Language), BSL (British Sign Language), and so forth, which are used primarily by deaf communities. Soon, however, it became clear that there are other distinct sociolinguistic types of sign languages, typically associated with smaller communities, in which hearing users play a larger role. One is *homesign*, in which a signing system emerges in households in which one or more children grow up in isolation from wider communities of signers (Goldin-Meadow 2003). Although the deaf children provide the raison d'être for the homesign, the household of homesign users typically contains more hearing users than deaf ones, though many of these may have limited mastery of the signing system. A second is *village sign language*, associated with typically esoteric communities in which, for a combination of cultural and genetic reasons, there is a significantly higher incidence of deafness, two well-documented examples being Al-Sayyid Bedouin Sign Language (Sandler et al 2005) and Kata Kolok (de Vos 2012). In such cases, the sign language is used not only by deaf people but also by a substantial proportion of the hearing people in the community. And a third is *nucleated network sign language,* in which, in typically rural areas, individual deaf persons live in isolation from one another, yet share a common sign language that is supported by its use amongst the general hearing population (Reed 2022). Such a situation obtains across much of Papua New Guinea and most probably in several other parts of the world.

Nevertheless, in all of the above cases, the development and use of sign languages is driven by the presence of deaf people, who need the sign languages in order to communicate. However, a handful of reported cases point towards the existence of sign languages whose primary function is not to facilitate communication amongst deaf people, and which instead are used primarily by hearing people; among these are the Plains Indian sign languages (West 1960) and the Aboriginal sign languages of the Australian North Central Desert (Kendon 1988). What is common to these cases is that they are all associated with pre-modern societies of a broadly esoteric profile. This paper focuses on what is

perhaps the most widespread case known so far of a sign language used by the general hearing community, the hitherto-undescribed *Kode Papua*.

2. Kode Papua

Kode Papua is used by most or all members of the general hearing population in at least some regions of Tanah Papua, the western half of the island of New Guinea, currently a part of Indonesia. Users of Kode Papua are typically fluent also in one or more oral languages, which provide their primary vehicle of communication; among these are Papuan Malay, the local lingua franca, and a large number of local languages, many of which are endangered.

Kode Papua is used as an alternative means of communication in a variety of contexts. First, it is the primary vehicle for communication between hearing and deaf people, the latter living largely in isolation from one another, and not familiar with other sign languages, such as the Indonesian sign language BISINDO. Significantly, however, the existence of Kode Papua is not predicated on the presence of deaf people; instead, it is used by the general hearing population in a variety of contexts in which no deaf people are present. Some of these contexts might involve situations in which oral communication is precluded, for example, communication at a distance, or in noisy environments. Alternatively, in a situation involving a group of people gathered together in oral communication, a subset of the group may break out and initiate a conversation in Kode Papua, so as not to disrupt the general ongoing discussion, or perhaps to convey a message that they wish to keep secret from the others. In other cases, though, the use of Kode Papua appears to be just for fun, such as during a silent game in which nobody is allowed to make a sound. Finally, Kode Papua is often used in situations of code switching with Papuan Malay or other oral languages, in which case the use of Kode Papua segues into a more familiar and widespread, probably universal phenomenon, namely the use of co-speech gestures.

Unlike co-speech gestures, however, Kode Papua bears many or most of the hallmarks of full-fledged languages, either signed or oral. First, its expressive power is considerably higher than that of co-speech gestures, and is instead on a par with that of other languages. Kode Papua can be used to refer to people, things, properties, states and activities, not only in the here and now but also in points distant in space and time; it can be used to gossip, joke, tell stories, negotiate transactions and relationships, and perform most of the communicative needs of everyday life. Grammatically, it falls within the general ballpark of other well-described sign languages, the differences between Kode Papua and other more familiar sign languages comparable to those observed amongst other sign

languages. Notwithstanding its exceptional sociolinguistic circumstances, preliminary investigations would seem to suggest that Kode Papua merits the characterization as a bona fide language.

3. A comprehension experiment

In order to confirm the use of Kode Papua amongst the general hearing population and to explore its geographical distribution, a comprehension experiment was conducted. Participants, all hearing (with one exception, noted below), were asked to view a short (1 minute 15 second) video showing selected passages of a naturalistic conversation between two users of Kode Papua, O and I, and then to describe, in their own words, what was said. Their descriptions were then coded in accordance with how much of the conversation they understood.

A transcription of the Kode Papua conversation is provided below:[1]

O: RECORD FOLLOW FOOTBALL 3SG RECORD
'Let's go and play football and he'll record us.'
I: HEY PHONE DROP NO.MORE FEAR
HEY NO.MORE PHONE\DROP NO.MORE FEAR NO.MORE NO.MORE
'I'm afraid they'll smash the phone to the ground and break it.'
O: HUH
'Huh?'
I: FEAR DROP NYONG DROP PHONE
'I'm afraid Nyong will smash the phone.'
COLLIDE BIKE UP UP BIKE COMPASSION HERE SLASH
'I crashed my bike and got hurt all over, they were slashing each other.'
HEY SWORD LONG SLASH UP RUN STAB HERE COMPASSION
'They were slashing each other with swords, running, stabbing.'

Participants' responses were coded in accordance with whether they correctly identified the presence of 12 meaning elements, as indicated below:[2]

- record
- football
- mobile phone
- drop
- [proper noun]
- no more
- bike
- collide
- sword
- slash
- run
- stab

[1] In the above transcription, HEY represents a sign consisting of an outstretched arm and open hand facing downwards, with a usually-repeated bending of the hand from the wrist or base of the fingers, downwards and back towards the signer; some of its numerous functions, illustrated in the above example, include beckoning, solicitation of attention, and the marking of turn or utterance onset (Inbar and Gil 2025). NYONG, formed by the fingers brought to the upper lip indicating a mustache, is a proper noun, referring to an individual whose father is renowned for his mustache.

[2] For "[proper noun]", participants, most of whom were unfamiliar with the person in question, were coded as successfully understanding the sign if they recognized it as referring to a particular person.

Each response was assigned a score of 1 if the meaning element was present, 0.5 if a meaning element related but not identical to the intended one was judged to be present, and 0 if the meaning element was entirely absent. Each individual participant was then assigned an average score over the 12 meaning elements, representing a measure of his or her comprehension of Kode Papua.

Participants were grouped into distinct populations according to geographic and sociolinguistic criteria. Some selected results are shown below:

Table 1. Kode Papua comprehension experiment: preliminary results.

	Sorong S	*Sorong*	*Wasior S*	*Wasior*	*Ternate*	*Semporna*	*Israel*	*Israel S*
n	16	14	12	9	14	13	12	10
AVG	0.69	0.65	0.64	0.60	0.38	0.09	0.19	0.21

In Table 1, columns represent distinct populations tested. The first row shows the number of participants in each population, while the second row contains the result that is of interest here, namely, the average score for all of the participants within the population in question.

In Table 1, Sorong is a major urban center on the western tip of New Guinea, in which the video was recorded. Wasior is a small town in a more rural setting in the south Cenderawasih Bay region. The remaining locations lie outside of Tanah Papua: Ternate in the Maluku islands of Indonesia, Semporna on the northeast coast of Borneo in Malaysia, and Israel as a control case in a completely different part of the world. For three of the locations, two distinct populations were tested. For Sorong, *Sorong S* refers to participants who were part of a social group containing a deaf person, who happened to be one of the two interlocutors in the video that functioned as the experimental stimulus, while plain *Sorong* refers to members of another social group who had no regular contact with any deaf person. Analogously, for Wasior, *Wasior S* refers to participants from the village of Rasiei, outside of town, who often communicated with a deaf person living in the village, while plain *Wasior* refers to people living in town who had little or no contact with any deaf person. Finally, for Israel, *Israel S* refers to deaf or hard-of-hearing persons familiar with ISL (Israeli Sign Language), while plain *Israel* refers to hearing people with no knowledge of any sign language.

The four populations from Tanah Papua all display scores in the 0.60-0.70 range, with no statistically significant differences between them. This shows that within the region in question, comprehension of Kode Papua is maintained across both geographical and sociolinguistic space. Geographically, Sorong and Wasior

are quite distant from each other, separated by several linguistic and cultural boundaries; no land transport is available between the two locations, the fastest connection being an 18-24 hour journey by large ship. The invariance of Kode Papua across such a large region, and perhaps even further, suggests that, at the very least, Kode Papua is not a recent development that emerged in the course of the last few generations. Sociolinguistically, the fact that, within Sorong, the friends of the two interlocutors in the video, one of whom was deaf, were not significantly better at understanding the conversation than another unrelated population, at 0.69 vs. 0.65, shows that Kode Papua is not some kind of homesign driven by the presence of one or more deaf persons; analogous observations obtain also with regard to the two populations in Wasior, at 0.64 vs. 0.60.

Outside of Tanah Papua, however, the degree of comprehension of Kode Papua drops off massively, to 0.09 for a population in Semporna, Malaysia, and 0.19 and 0.21 for two populations in Israel. This suggests that the successful understanding of the experimental stimulus in Tanah Papua was due, at least in part, to arbitrary, conventionalized properties specific to Kode Papua, and not just universal principles of iconicity. Or in other words, that Kode Papua is a real language. Further support for this conclusion comes from the fact that in Israel, knowing another unrelated sign language, namely ISL, did not help participants understand Kode Papua.

Finally, in Table 1, an intermediate position is occupied by Ternate, with its score of 0.38. On the one hand, this score is significantly lower than those of the four Tanah Papua populations, suggesting that Ternate lies outside the geographical boundaries of Kode Papua. On the other hand, the Ternate score is significantly higher than those of more geographically distant populations, which could be due either to the presence of some other related sign language, or to the use of co-speech gestures similar to those that form part of Kode Papua.

4. Conclusion

As argued here, the use of Kode Papua and other similar sign languages by substantial populations of hearing people in esoteric societies around the world suggests that the use of sign languages amongst the general hearing population was more widespread at an earlier stage in the evolution of language. This conclusion supports the characterization of co-speech gestures around the world as evolutionary relics from an earlier stage in the evolution of language, and, more generally, is consistent with the gestural theory of language evolution (Corballis 2009, McNeill 2012, Gillespie-Lynch 2014, Prieur 2020).

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